Folkliga fanor i Uppland

Digitalisering för bevarande och tillgänglighet
Högskolan i Borås
Vårterminen 2026.

Jenny Aspelund
Cecilia Pulkkanen
Maria Store

Contents

Introduction 2
- Folkrörelsearkivet 2
- Purpose of digitising the banner collection 2
- Historical background of the banners 3

Methodology 3
- Image capture 3
- Image editing 5
- TEI - Text Encoding Initiative 5
- Publication 6

Discussion 6
- Why do we digitise? 6
- The historical value of banners 7
- Digital images 8
- TEI 8
- Copyright 9
- Closing reflections 9

References 10

Appendices 11
- Appendix A 11
- Appendix B 12

Introduction

The task in this project was to digitise a limited amount of cultural heritage objects in collaboration with a cultural heritage institution. We reached out to Folkrörelsearkivet för Uppsala län (the Popular Movements Archive in Uppsala), and were offered to participate in a project of digitising popular movement banners in preparation for a lecture and an exhibition on the subject that was planned for April of 2026.

Folkrörelsearkivet

Folkrörelsearkivet för Uppsala län was established in 1978 and serves to collect, preserve and store historical information about popular movements. It is mainly financed by the municipalities of the county and through government grants. The archive, which is housed in rather small facilities next to the public library in Uppsala, holds documents as well as photographs, posters, objects, recordings and banners. The staff aim to make the material accessible by organising exhibitions, lectures, field visits and courses. Due to limited resources, much of the archive's material remains unsorted and in many cases the staff don't know exactly what their collections contain, or where a certain object is located.

Archive material

Purpose of digitising the banner collection

The banner collection of Folkrörelsearkivet consists of roughly 385 banners from the 19th and 20th century, together with accessories such as rods, top emblems and lacings, but also documentation about their origin such as protocols recording their purchase or articles about their use.[1] Banners have been donated continuously since the founding of the archive. In the 1990s, an index was created for the banners, but this is rather makeshift in character, with very small digital images and incomprehensive data.

There are plans to move and store the banners in a facility located outside of Uppsala, which would make it even more difficult to access the banners. This, together with the fact that the banners are old and sometimes fragile and need to be protected, has influenced the decision to digitise the banners. The process has been further hastened by the need of good

[1] Some of the information in this section has been conveyed by head archivist Örjan Simonsson at Folkrörelsearkivet.

photographs of the banners for the exhibition and lecture of writer and researcher Margareta Ståhl.

Historical background of the banners

Traditionally, the banner of a popular movements association was considered almost sacred and it was the most important possession of, for example, a trade union. It was a symbol of the association and a means to unite the group. At the same time, the banner was a message directed outwards whenever it was used in demonstrations, celebrations and, sometimes, funerals. The earliest banners were made of silk, which was very expensive, and an association usually only owned a single banner. It could take a long time to raise the money needed to buy a banner, and once the goal was achieved the banner was used until it was worn out. Later on, less fragile materials like wool and synthetic fabric were used for the banners.

The banners of the liberal movements were generally white in the beginning, but from 1866, when the First Socialist International was held in Switzerland, the red banner became the symbol of the labour movement. The first red banner in Sweden appeared in 1883, and by the first May Day demonstrations in 1890 there were at least 70. As a symbol of power, war, love, brotherhood, equality, liberty and justice, the red banners caused great commotion and were often seen as aggressive and provocative.
Most of the banners at Folkrörelsearkivet are red and associated with the social democratic labor movement. Many of them are hand painted with motifs chosen by the ordering association and crafted by a banner artist who worked in one of several large banner studios (fanateljéer) that emerged in the 19th century in the wake of the growing Swedish popular movements. Women associations often made their banners themselves and in that case they were usually embroidered and not painted, but still artistic and thoroughly manufactured; it could take up to a year to finish one banner this way.

A red banner

Methodology

Image capture

The selection of which banners to digitise was made by Margareta Ståhl, who wished to include 30 different banners in her lecture. This meant that we couldn't exert much influence on the choice of material. Since the banners are of considerable size, as well as old and fragile, we needed to photograph them on the premises of the archive. An advantage was that we were allowed to use the in-house equipment, and on our first session of image capture we were instructed by two interns who had started photographing and tried out settings for the camera. The photoshoot took place in an office room/library with windows only partly covered by blackout curtains, which was a possible source of distortion because of changing light during the day. The floor had been cleared and a black canvas was used as an underlay for the banners. The camera, a Nikon Z 6 with a 35 mm photo lens, was attached to the ceiling. There were two upward lights with a colour temperature of 5600 Kelvin, which is the tone of natural daylight.

A big part of the work consisted in finding and handling the banners. They are kept in drawers in a high metal cabinet in the basement, or rolled up on cardboard rolls placed on top of filing cabinets in the archive. It took a great deal of time to find them, fold them with care, transport them to the photo room, unfold them on the floor and arrange them as smoothly as possible. It was also time-consuming to return them to their location since we needed a ladder in order to put them back in the right position. With time we developed a better strategy and learned how to handle the banners more efficiently..

Storage of banners

Once we had the banner ready for photographing, we could see it on the computer screen in the program Capture One Pro. We didn’t have access to colour cards, and it was not easy to get a good presentation of the colours. On a couple of occasions we couldn’t get the exact same colour on both sides of the banner, even though there were no obvious changes in light or other conditions. We adjusted the settings, but in the end we had to settle for the best possible result, even if it wasn’t perfect. Since the camera was attached to the ceiling it was not entirely easy to handle. The photo could be taken remotely, but to change the battery or adjust focus we needed a ladder. The photos were taken in RAW and exported as TIFF and JPEG to GIMP where we could crop the image and save it. The resolution is 300 ppi and the colour depth is 8 bits per channel.

Cowick (2018, p. 92) states that 600 ppi and 8 bits is a minimum requirement when digitising paintings and other two-dimensional works of art, but the archive staff had decided on 300 ppi, even though they were considering the possibility of using a higher quality in the future. We came to the archive premises on three occasions and photographed nine banners on both sides, a total of eighteen images.

Banner ready to photograph

Image editing

The images were stored on a cloud at Folkrörelsearkivet, where we were able to access them after our image capture sessions. They were then edited using Adobe Photoshop. The overall aim of this editing was to achieve a picture that did justice to the actual artefact that we had handled at the archive and seen with our own eyes, while keeping the changes as subtle as possible.

The original TIFF files were saved locally as well as uploaded to a cloud. They were then exported to JPG format for use on the webpage. Some metadata was embedded in the image files using the metadata section in Photoshop. In the IPTC field, short descriptions were added stating which organisation the banner represented, when and where it had been made (if known) and where it is stored. The images were named in accordance with the guidelines presented in Cowick (2018), which state that file names should be “clear and concise”, contain “sufficiently descriptive information”, and use “hyphens instead of periods, underscores or spaces” (p. 37).

TEI - Text Encoding Initiative

Vanhoutte (2004, p. 9–10) describes the search for a system to represent computer readable data as an aim for every project in the humanities since the first uses of computers and how every different scholar or research group tended to create their own system. The result was a multitude of encoding schemes, and there was obviously a need for a common strategy. The Text Encoding Initiative (TEI) means to “develop and maintain guidelines for the digital encoding of literary and linguistic texts” (TEI, 2026) and provides the TEI Guidelines with some 600 elements that can be used in encoding a variety of texts in any language. TEI offers recommendations rather than a standard for encoding, which makes it possible to express individual theories by encoding features according to one’s own opinion of what is important in a text (Vanhoutte, 2004, p. 11). This openness to interpretation also made it possible for us to use TEI even though our focus was not on text primarily, but on images and the description of objects.

We created a <teiHeader> with the metadata common to all banners assembled, both administrative, such as information about availability according to a Creative Commons license, and descriptive. Then we used the <listObject> element and described every banner in a separate <object> element. The <object> element is a recent addition to the guidelines and is described as “a description of a single identifiable physical object” (TEI, 2026, 14.3.6).

With the help of the guidelines it was possible to find elements to describe the banners in a fairly detailed manner; material, colour, size, symbols, text and origin. With more time it would have been possible to add even more details by using for example the element <condition> to describe discoloration, repair, or other signs of usage. In the <object> element we also added information about where the physical banner is located and the id number used by Folkrörelsearkivet, to facilitate for someone who would wish to gain access to, or information about, the actual banner.

Publication

An important part of availability is the publication of the material. In our project, the material about the banners was published on a webpage made from templates provided in the course, through GitHub, which supplies a repository. Since we had made the choice to use TEI in our work it would perhaps have been natural to use XSLT in order to transform the TEI-XML file into HTML format for the webpage, but we decided not to for two reasons; firstly it gave us the possibility to work with the webgage and the TEI file in parallel and, secondly, we considered the effort of learning to use XSLT challenging. This is not a recommended solution with a more extensive body of material, since working on the HTML file manually could be tedious work.

The webpage has four tabs; one home tab with general information about the project and one gallery tab with all the banners, with clickable images to show the backside of the banner, a short informative text and metadata concerning each banner. The metadata includes title, creator and size as well as text and symbols on both sides of the banner. The third tab contains information about the historical significance and usage of the banners and is based on an interview, a lecture and the writings of Margareta Ståhl. The final tab has additional information about the group members, the people who have contributed in different ways, and links to our material in GitHub.

The images in this project are distributed under a CC-BY 4.0 license which enables reusers to freely adapt or share the material as long as the creator is given appropriate credit. This information is embedded in the image file metadata and in the TEI file.

Discussion

Why do we digitise?

There are several reasons for digitising cultural heritage objects. Most of them have to do with increased access, preservation, or minimizing the handling of fragile objects. A Swedish government investigation even describes digitisation as a prerequisite for democratic access to cultural heritage (Kjellman, 2009, s. 219).

At the same time, it is important to remember that the digital representation of an analogue object is an interpretation and not a replacement of the original object itself (Terras, 2015, p. 67). Björk (2015) argues that “the actual shape and structural composition of the source document […] might be rendered as an image”, but adds that “the material presence of the document, e.g. touch, smell, and weight, is not as readily conveyed” (p. 203). This is very obvious in relation to banners, where all the physical features such as the size and weight, as well as the different textures and colours, are reduced to a two-dimensional image.

Another problematic aspect concerns the swiftly changing digital landscape, which makes it difficult to ensure that seemingly long-lasting solutions will in fact last. In many ways, digitisation is not as much of an enduring solution to the preservation aspect as we would like to believe. A couple of years ago, a survey directed to several Swedish memory institutions showed that most of them only had a strategy for storing and preserving digital archives on a short-term basis of about five years, and that none of them had a strategy for long-term preservation (Digisam, 2014, p. 12). The distinction between “storage” and “preservation” is an important one, as the latter prescribes that data is secured in such a way that future users will be able to not only find the information, but also understand and absorb it (p. 10).

In the case of the banners at Folkrörelsearkivet, increased access would seem to be the strongest argument for digitising. In its current state, the collection is almost entirely invisible to the public. If it were to be made accessible, with high-resolution images and relevant descriptions as well as metadata, potential viewers would at least get some idea of the appearance and unique historical value of the banners in question.

The historical value of banners

In her dissertation on union banners as a means of visual communication, Ståhl (1999) points out that the use of banners in public demonstrations changed in the 1880s, when the first red ones appeared in Sweden: “Fanans defensiva funktion att stärka gemenskapen övergick till offensiv politisk kamp, från samlingstecken till kampfana” (p. 313). She also maintains that elaborately crafted banners could be regarded simultaneously as “working class lore”, and as a kind of “visual agitation” (p. 316, 318). Their foremost value lies in their ability to convey a message: “Med hjälp av den röda fanan erövrade arbetarrörelsen […] sin del av den samhälleliga arenan både konkret och symboliskt” (p. 318).

Every banner is a symbol of the local organisation it represents, as well as the larger labour movement. As such, they contribute to knowledge about the rites and traditions of the people involved in these movements.

Prescott and Hughes (2018) have pointed out that one dominant trend in the cultural heritage sector has been “an emphasis on the digitisation of ‘treasures‘“ (p. 5). In their view, this threatens to reinforce canonisation and ’long-standing cultural narratives” (p. 6). Digitising union banners, or any of the artefacts kept by popular movements archives, seems like an obvious contrast to this trend. Prescott and Hughes also make a case for “slow digitisation”, which indeed seems like the only way to go about working with banners and similar objects. In this aspect, the process can be said to conform to Dahlström’s (2011) concept of “critical digitisation”, as opposed to “mass digitisation”. According to Dahlström, the critical digitisation process normally involves having to develop “contingent procedures and tools and tailor them to the nature of the documents in the particular collection”, (p. 97), which certainly was the case in this project.

Digital images

According to Terras (2008, p. 40), the resolution of a digitised image depends on its purpose. In some circumstances you know which quality is required, for example if you are publishing a newspaper, but the future use of a digital image is not always known. Consequently, the best procedure in handling rare and fragile materials would be to take one image with high resolution, to use for different purposes, and thus protect the source material. In our project, the choice of resolution was made by the staff of the archive and it was obvious that they did not know exactly what would be the best level of quality, partly because they were not sure how the images would be used in the future. The immediate need for the pictures was connected to a lecture where they would be shown using a projector, together with an exhibition with printed versions of the files. This meant that the staff were interested not only in the resolution in ppi (pint per inch), but also in dpi (dots per inch) regarding the printing quality. They settled for 300 ppi and 2400 dpi even though, for example, Cowick (2018, p. 92) recommends at least 600 dpi when digitising two-dimensional works of art, which is what you could call the banners.

The image editing presented its own challenges. As we had worked in a room where we were not able to uphold perfect lighting conditions, or even similar conditions for every single photograph taken, the images showed slight variations in lightness and colour. To compensate for these flaws, gentle adjustments were made using tools such as “levels”, “lightness/contrast” and “hue/saturation”. Visible dust and dirt from the image capture was removed using the “clone stamp” tool. The banners having been photographed on the floor, on a black curtain which we also had to walk on to adjust each object before taking the picture, there was quite a lot of work to be done. Again, the aim was to represent the banner in the best way possible, without distracting elements. Any stains, specks, tears or discolourings on the actual fabric were of course left untouched in the editing process.

Interestingly, Björk (2015) points out that in some cases, “the digital presentation can also be used to emphasise aspects that are not as discernible when engaging with the ‘physical’ document” (p. 203), due to the fact that a digital image can be enlarged to show things in greater detail, such as a fingerprint on a manuscript. The banners display many signs of wear and tear, including patchings and mendings, which could potentially be studied more closely in a high-resolution image. They may be regarded as an integral part of the artefact and its story.

TEI

Since the original aim of the Text Encoding Initiative was to provide a means to digitally encoding texts it is natural that the TEI Guidelines focuses on this aspect and that there are a lot of examples and literature dealing with text encoding and TEI. TEI has also become more or less a standard in text encoding and, for example, the DFG Practical Guidelines (Deutsche 10 Forschungsgemeinschaft, 2013, p. 33), demands that digitized full text be encoded and marked up according to TEI unless there are very good reasons not to. They also state that TEI is the best choice for long term archiving.

In our project we have been working primarily with images and not text, which means that we had to decide whether to use TEI or find another solution. Dappert et al. (2016, p. 3) emphasise the importance of using standards both as a way of learning from others and thus reducing the risk of overlooking important metadata in your own work, but also as a way to facilitate the exchange of information between different actors. Both these factors and the recommendation of using TEI for long term archiving made us willing to examine the potential of TEI for our project, but since it was difficult to find literature or examples of projects similar to ours, we had to rely mostly on the TEI Guidelines to find out if it was possible to describe our banners in an appropriate way using the TEI elements.

As it turned out the guidelines had a new element, the <object> element, which suited our purposes since it is used to describe “any material thing whether real, in existence, fictional, missing, or purported about which more information is known” (TEI, 2026, 14.3.6). A possible complication, though, in using a new element is the risk that it will develop and be revised with time, in which case our file would need to be adapted to these changes.

Copyright

In the Swedish Copyright Act for Literary and Artistic Works (SFS 1960:729, 4 Kap 43-44 §) it is declared that the right to a work is protected until 70 years after the death of the author or creator or, if the originator of the work is unknown, until 70 years after the work was made public. These legal issues are one of the considerations you need to take into account when making decisions about digitisation, since the legal status of a work is not always obvious and can be difficult and time-consuming to determine (Oruç, 2025, p. 102). This was partly the case with the banners in our project. Most of them are old enough to clearly belong to the public domain, but others are more recent or have an unclear date of creation and furthermore the creator may be unknown. To focus on public domain is helpful for avoiding legal risks, but can also cause a situation where only part of a collection is digitised to the annoyance of future researchers (Oruç, 2025, p.102). In our case we did not have the time to find out ourselves about the history of the separate banners, but fortunately we were able to get expert help from the staff of the archive and also from Margareta Ståhl when deciding whether the banners could be digitised and published without problems.

Closing reflections

This project has taught us many things about the digitisation process and all its different aspects. One key takeaway is that there is a lot to be done before the actual work begins. Apparently, it is not unusual that the time required for preparatory activities in digitisation projects are underestimated (Deutsche Forschungsgemeinschaft, 2013, p. 12). We had a fairly good idea how we were going to use the material involved in our own project, but a long-term solution for the entire banner collection would have required much more thorough preparations.

Each time we photographed an artefact we learned something new, and eventually this knowledge helped us work more effectively. Even so, we estimate that the image capture would require at least one hour per banner. The time spent on image editing could be reduced if the objects were photographed under better conditions regarding lighting and backdrop, but some work would still be necessary. With time added for metadata processing and web publication, we would be looking at a couple of hours in total for each banner. At a rate of three hours per banner, for instance, the entire process would entail more than a thousand hours.

As of May 2026, the future destiny of the banners at Folkrörelsearkivet is still uncertain. A new storage solution has been proposed, but it is costly and would call for additional funds. In all probability, the banners will eventually be moved and the physical objects will be represented by high resolution images. Head archivist Örjan Simonsson (personal communication, April 28, 2026) also believes that the current index would need to be assessed by an expert, as some of the information in the current register is inaccurate. All in all, it is going to take a lot of work to achieve a digital collection of good quality.

Our contacts with the staff at Folkrörelsarkivet, and their generosity in letting us access their equipment and collections, gave us valuable insight into the actual challenges that many memory institutions face. The fact that they often have to deviate from best practices, because of scant resources or practical concerns, was another important lesson learned in this project.

References

Björk, L. (2015). *How reproductive is a reproduction? Digital transmission of text-based documents*. University of Borås.

Cowick, C. (2018). *Digital curations projects made easy : a step-by-step guide for libraries, archives, and museums*. Rowman & Littlefield.

Dahlström, M. (2011). Editing Libraries. In C. Fritze, F. Fischer, P. Sahle & M. Rehbein, *Bibliothek und Wissenschaft. Vol. 44: Digitale Edition und Forschungsbibliothek*. Harrassowitz. 91-106.

Dappert, A., Guenther, R. S., & Peyrard S. (Eds.). (2016). *Digital preservation metadata for practitioners: Implementing PREMIS*. Springer.

Deutsche Forschungsgemeinschaft (2013). *Practical Guidelines on Digitisation*.

Digisam (2014). *Digitalt bevarande vid kulturarvsinstitutioner. Nulägesanalys och framtida behov*. Digisam.

Kjellman, U. (2009). Digitalisering som standardisering: verktyg i koncipieringen av kulturarv. I N.D. Lund (Red.), *Digital formidling af kulturarv: Fra samling til sampling.* Multivers.

Oruç, P. (2025). *Digitising Cultural Heritage: Clashes with Copyright Law*. Hart Publishing.

Prescott, A., & Hughes, L. (2018). Why Do We Digitize? The Case for Slow Digitization. *Archive Journal,* September 2018, https://www.archivejournal.net/essays/why-do-we-digitize-the-case-for-slow-digitization/

SFS 1960:729. Lag om upphovsrätt till litterära och konstnärliga verk. https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/lag-1960729-om-upphovsratt-till-litterara-och_sfs-1960-729/#K4

Ståhl, M.. (1999). *Vår fana röd till färgen. Fanor som medium för visuell kommunikation under arbetarrörelsens genombrottstid i Sverige fram till 1890*. Linköpings universitet.

TEI: Guidelines for Electronic Text Encoding and Interchange (2026-04-08). https://tei-c.org/release/doc/tei-p5-doc/en/html/index.html

Terras, M. (2008). *Digital Images for the Information Professional*. Ashgate.

Terras, M. (2015). Cultural heritage information: artefacts and digitization technologies. In I. Ruthven & G.G. Chowdury (Red.), *Cultural heritage information: access and management* (p. 63–88). Facet.

Vanhoutte, E. (2004). An Introduction to the TEI and the TEI Consortium, *Literary and Linguistic Computing*, *19*(1), 9–16. https://doi-org.lib.costello.pub.hb.se/10.1093/llc/19.1.9

Appendices

Appendix A

Activities project work

Week	**Activities**
5	Planning: group meeting, mail contact and meeting Folkrörelsearkivet
6	Preparations: Reading of literature
7	Preparations: Reading of literature
8	Preparations: Reading of literature
9	Preparations: Reading of literature
10	Preparations: Reading of literature, visit to Folkrörelsearkivet
11	Preparations: Lectures and workshops in Borås, meeting with advisor
12	Image capture
13	Image capture, group meeting
14	Meeting with Margareta Ståhl, group meeting, work with images, metadata, TEI, web page, report writing
15	Group meeting, work with images, metadata, TEI, web page, report writing
16	Group meeting, work with images, metadata, TEI, web page, report writing, attending lecture on banners
17	Work with images, metadata, TEI, web page, report writing, meeting with advisor
18	Group meeting, web page, report writing

19	Group meeting, web page, report writing
20	Group meeting, web page, report writing
21	Web page, report writing, presentation of work in seminar
22	Submission of work

Appendix B

Work time*

Activity	Time
Group meetings	25 h
Other meetings, mail contacts	22 h
Image capture	37 h
Image editing	38 h
TEI-XLM	35 h
Web page, GitHub	52 h
Report writing	50 h
Total:	259 h

* Reading of literature is not included since it was difficult to distinguish from the other activities and the rest of the course.